

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA0304 COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO1:** Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

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QUESTIONS: 10

Total Marks: 50

Annexure A

Case Study

Duration: 2hrs

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. **(5 Marks)**
2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**
3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining

states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**

4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**
5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**
6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**
7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance. **(5 Marks)**
8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. **(5 Marks)**
9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. **(5 Marks)**
10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design

a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. **(5 Marks)**

Code of Conduct:

I K Varshan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CASESTUDY

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Answers

1. Warehouse Robot (Pick, Pack, Transport)

MDP Components

- **States:** Robot's (x, y) position/orientation, shelf/item locations, current payload status (empty/carrying item), battery level, positions of obstacles/other robots, target pick/drop location.
- **Actions:** Move (forward/backward/left/right), rotate, pick item, place item, wait/stop.
- **Environment:** Warehouse grid/map with shelves, dynamic obstacles (other robots, staff), item locations, docking/packing stations.
- **Policy:** Maps current state (position + task status) to the next best action to reach the target efficiently and safely.

Reward Function Design

- +R_task on successful pick/pack/delivery (large positive, scaled by speed).
- Small negative step penalty (e.g., -1 per timestep) to encourage speed.
- Large negative penalty for collisions or safety-zone violations (e.g., -50).

- Small penalty for dropped/misplaced items (accuracy).
- Bonus for completing task under a time threshold.

Example: $R = R_{\text{delivery}} - \alpha \cdot (\text{time_steps}) - \beta \cdot (\text{collision}) - \gamma \cdot (\text{item_error})$

Analysis/Evaluation

- If speed is over-weighted, the robot may take unsafe shortcuts near obstacles, increasing collisions.
 - If safety penalties dominate, the robot becomes overly cautious, slowing throughput.
 - Balanced weighting (α , β , γ tuned via experimentation) is essential — reward shaping directly determines whether the learned policy is fast-but-reckless or safe-but-slow. Sparse rewards (only at delivery) slow learning; intermediate shaping rewards (progress toward shelf) speed convergence.
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2. Movie Recommendation on a Streaming Platform

MDP Components

- **State:** User profile embedding, recent viewing history, genre preferences, time of day/session context, watch duration of last recommendations.
- **Action:** Recommend a movie/show from the catalog (or a slate of items).
- **Environment:** The user, whose feedback (watch, skip, rate, abandon) is observed after each recommendation.

- **Reward:** Positive for watch completion/high watch-time or explicit rating; negative/zero for skip or quick abandonment.

Reward Function Design $R = w_1 \cdot (\text{watch_time_ratio}) + w_2 \cdot (\text{explicit_rating}) + w_3 \cdot (\text{click}) - w_4 \cdot (\text{skip})$

Analysis/Evaluation

- Reward based purely on clicks encourages clickbait-like recommendations (short-term engagement) but hurts long-term satisfaction.
 - Reward based on watch-completion promotes genuinely relevant content but may narrow recommendations to "safe" choices.
 - **Exploration vs. Exploitation trade-off:** Exploitation (recommending known-liked genres) maximizes immediate engagement; exploration (recommending new genres/content) is needed to discover latent preferences and avoid a filter bubble. An ϵ -greedy or UCB-style strategy balances discovering new preferences against maximizing immediate watch-time.
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3. Autonomous Irrigation & Fertilizer System

MDP Components

- **States:** Soil moisture, soil nutrient levels, weather forecast (rain probability, temperature), crop growth stage, historical yield data.
- **Actions:** Irrigate (amount/duration), apply fertilizer (type/amount), do nothing.
- **Environment:** Field/farm with soil and crop dynamics, influenced by weather.

- **Reward:** Based on crop health indicators and, ultimately, yield/quality at harvest, with penalties for water/fertilizer overuse.

Reward Function Design $R = R_yield_gain - c1 \cdot (water_used) - c2 \cdot (fertilizer_used) - c3 \cdot (crop_stress_penalty)$

Analysis/Evaluation (Delayed Rewards)

- The true reward (yield) is only known at harvest — weeks/months after actions are taken, making credit assignment difficult (which irrigation decision contributed most to yield?).
 - This delay favors algorithms that handle long-horizon credit assignment well, such as **Temporal Difference (TD) learning or Deep Q-Networks with discounted future rewards**, over pure Monte Carlo methods, since TD learning updates values incrementally without waiting for episode completion.
 - Intermediate proxy rewards (e.g., soil moisture staying in optimal range) can be added to speed up learning while still ultimately optimizing for yield.
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4. Cybersecurity Intrusion Detection & Response

MDP Components

- **States:** Network traffic features (packet rate, protocol type, connection patterns), system logs, anomaly scores, current threat level.
- **Environment:** The live network, including both normal traffic and potential attackers.
- **Actions:** Allow traffic, block IP, flag for review, throttle connection, isolate host.

- **Reward:** Positive for correctly identifying/blocking an attack; strongly negative for false positives (blocking legitimate users) and false negatives (missed attacks).

Reward Function Design $R = +R_{\text{true_positive}} - p1 \cdot (\text{false_positive}) - p2 \cdot (\text{false_negative}, \text{weighted by severity})$

Analysis/Evaluation

- **Sparse rewards:** Actual attacks are rare relative to normal traffic, so meaningful positive/negative reward signals occur infrequently, slowing learning — this is compounded by class imbalance.
 - **Real-time constraints:** The agent must decide within milliseconds, limiting the use of computationally heavy policies; lightweight function approximators and pre-trained/offline RL are often used with fast online fine-tuning.
 - Mitigations include reward shaping with intermediate anomaly-score-based rewards and prioritized experience replay to make rare attack events more influential during training.
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5. Ride-Sharing: Driver Allocation & Dynamic Pricing

MDP Components

- **States:** Current driver locations, rider demand distribution (by zone/time), traffic conditions, time of day, current price multipliers.
- **Actions:** Assign/reposition drivers to zones, set price surge multiplier for a zone.
- **Environment:** City road network with stochastic rider demand and driver supply.

- **Reward:** Revenue generated, minus penalties for long rider wait times or driver idle time.

Reward Function Design $R = \text{revenue} - \lambda_1 \cdot (\text{rider_wait_time}) - \lambda_2 \cdot (\text{driver_idle_time})$

Analysis/Evaluation

- **Environmental uncertainty:** Demand is stochastic and affected by weather, events, and time — the agent must learn a policy robust to this variability, favoring algorithms that model uncertainty (e.g., stochastic policies, robust MDPs) over purely deterministic ones.
- **Ethical concerns:** Purely profit-maximizing reward functions can lead to excessive surge pricing during emergencies or in underserved areas, raising fairness concerns. Reward functions should incorporate fairness/price-cap constraints, and pricing policies may need regulatory oversight and transparency.